

AI-Powered Smart Energy Consumption Prediction System

A Project Report

Submitted

In partial fulfillment of the requirements for the award of the degree



BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE and ENGINEERING

By

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that the report entitled “**AI-Powered Smart Energy Consumption Prediction System**” has been submitted by **Ganpisetty Maneesha (221FA04151)** and **Kolasani Kumari Supraja (221FA04161)** in partial fulfillment of the requirements for the Major Project course, as part of the academic curriculum of the **B.Tech. CSE Program**, Department of Computer Science and Engineering (CSE) at **VIGNAN’S FOUNDATION FOR SCIENCE TECHNOLOGY AND RESEARCH (Deemed to be University)**.

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DECLARATION

We hereby declare that the Project report entitled “**AI-Powered Smart Energy Consumption Prediction System**” submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech) in Computer Science and Engineering at VIGNAN’S FOUNDATION FOR SCIENCE TECHNOLOGY AND RESEARCH, Deemed to be University** is a record of our original work. This project has been carried out under the supervision of **Mr. Sourav Mondal** in the Department of Computer Science and Engineering, **VIGNAN’S FOUNDATION FOR SCIENCE TECHNOLOGY AND RESEARCH (Deemed to be University)**. The work embodied in this project has not been submitted previously, in part or full, to any other University or Institution for the award of any degree or diploma.

We have duly acknowledged all sources of information and data used in the preparation of this Project report and shall abide by the principles of academic integrity and ethical guidelines.

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ABSTRACT

Accurate prediction of energy consumption is essential for improving energy efficiency, optimizing resource utilization, and ensuring sustainability in smart city environments. With the rapid growth of smart buildings, IoT-enabled devices, and intelligent energy systems, effective energy demand forecasting has become increasingly important for reducing operational costs, minimizing energy wastage, and enabling better decision-making in energy management. The motivation behind this study is to develop an intelligent and reliable energy consumption prediction system that can accurately estimate future energy usage while providing insights into the key factors influencing consumption. To achieve this, a comprehensive methodology combining machine learning and deep learning techniques is proposed using real-world smart city sensor data, including air conditioning load, lighting load, plug load, temperature, humidity, ambient light, and temporal features such as year, month, day, and hour. Machine learning models such as Ridge Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and XGBoost, along with deep learning models including Artificial Neural Networks (ANN), LSTM, and a Hybrid Transformer–LSTM model, are implemented and evaluated. Experimental results demonstrate that among all the models, LSTM emerged as the best-performing model, effectively capturing complex temporal dependencies and energy usage patterns, while the Hybrid Transformer–LSTM model also contributed to improved predictive analysis. Ensemble models such as Random Forest and XGBoost delivered strong performance, further validating the effectiveness of the proposed framework. Additionally, SHAP (SHapley Additive exPlanations) analysis was integrated to enhance model interpretability, identifying temporal features, plug load, and air-conditioning load as the most influential factors affecting energy consumption. The proposed system provides accurate, reliable, and explainable energy forecasts, supporting efficient load balancing, intelligent energy optimization, and sustainable smart energy management in smart city infrastructures.

Keywords – Energy Consumption Prediction System, Machine Learning, Deep Learning, LSTM, Hybrid Transformer–LSTM, Random Forest, XGBoost, SHAP Analysis, Smart Energy Management.

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CHAPTER-1

INTRODUCTION

1.1 Energy Consumption Prediction and Its Importance in Smart Cities

Energy consumption prediction is the process of estimating future energy usage based on historical data and various influencing factors. It plays a vital role in efficient energy planning, resource optimization, and sustainable development, particularly in smart cities, smart buildings, and industrial environments where energy demand changes continuously. Accurate prediction of energy consumption helps reduce energy wastage, optimize overall usage, and ensure a stable and reliable power supply. With the rapid growth of urbanization, increasing dependence on electrical systems, and rising environmental concerns, intelligent energy forecasting has become essential for modern energy management. It supports better decision-making and helps organizations improve operational efficiency while reducing costs and environmental impact.

At its core, energy consumption prediction involves analyzing historical energy usage patterns and identifying the various factors that influence demand. Several load-based parameters, such as air conditioning load, lighting load, and plug load, directly contribute to the total energy consumed within a system. Environmental conditions such as temperature, humidity, and ambient light also play a major role in determining energy requirements. For example, higher temperatures often increase cooling demand, while lower natural light levels lead to increased use of artificial lighting. These factors continuously affect energy usage and make prediction a complex but necessary task. Understanding the relationship between these parameters improves forecasting accuracy and enables better control of available energy resources.

Time-based features such as the hour of the day, day of the week, month, and seasonal variations significantly impact energy consumption patterns. Energy demand often changes according to human activities, occupancy behavior, and operational schedules. Peak hours generally experience higher energy usage due to increased appliance and equipment operation, while off-peak periods show reduced demand. Human behavior, inefficient appliance usage, and lack of energy awareness can further contribute to unnecessary energy wastage. By analyzing these temporal and behavioral patterns, intelligent systems can identify recurring trends and improve the precision of energy predictions. This information is essential for efficient load balancing, energy planning, and optimized power distribution.

Accurate energy consumption prediction offers numerous benefits for both economic and environmental sustainability. It helps reduce operational costs, improve energy efficiency, and support better allocation of resources in smart energy systems. In smart grids and intelligent buildings, predictive models can prevent energy overloads, minimize power shortages, and enable proactive energy management. Additionally, improved forecasting contributes to environmental conservation by reducing excessive energy generation and lowering greenhouse gas emissions. Overall, energy consumption

prediction is a fundamental component of intelligent energy management and plays a crucial role in developing reliable, efficient, and sustainable smart city solutions.

1.2 Motivation

The increasing demand for energy due to rapid urbanization, population growth, and technological advancements has made efficient energy management a critical challenge in modern society. Smart buildings, industries, and urban infrastructures consume large amounts of electricity every day, making it essential to optimize energy usage and reduce wastage. Traditional energy management methods often fail to accurately predict future energy demand, leading to inefficient resource utilization, higher operational costs, and unnecessary environmental impact. These challenges highlight the urgent need for intelligent systems capable of forecasting energy consumption accurately and supporting better decision-making.

Another major motivation for this project is the growing concern over environmental sustainability and climate change. Excessive energy consumption, especially from non-renewable energy sources, contributes significantly to greenhouse gas emissions and environmental degradation. By predicting energy demand more accurately, organizations can optimize energy distribution, reduce unnecessary power generation, and improve the integration of renewable energy resources. Intelligent energy prediction systems can therefore play an important role in supporting sustainable development and promoting environmentally responsible energy management practices.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have opened new opportunities for improving energy forecasting systems. These technologies can analyze large volumes of historical and real-time data, identify complex patterns, and generate highly accurate predictions. However, many existing systems still face limitations related to prediction accuracy, scalability, and interpretability. This motivates the development of a more advanced and reliable energy prediction model that combines multiple ML and DL techniques with explainable AI methods to improve both performance and transparency.

The motivation behind this project is to design an intelligent and efficient AI-Powered Smart Energy Consumption Prediction System that can support smart buildings, smart grids, and future smart city infrastructures. By leveraging advanced prediction models and explainability techniques, the proposed system aims to optimize energy usage, reduce operational costs, improve energy efficiency, and contribute toward sustainable energy management. This project addresses real-world challenges and supports the development of smarter, greener, and more reliable energy systems.

1.3 Problem Definition and Research Gaps

Accurate energy consumption prediction remains a significant challenge in modern smart energy management systems due to the dynamic and complex nature of energy usage patterns. Energy consumption is influenced by multiple factors such as environmental conditions, load-based parameters, temporal variations, and human behavior, making it difficult to develop highly reliable forecasting models. Traditional statistical and machine learning approaches often struggle to capture these complex relationships effectively, leading to reduced prediction accuracy and limited adaptability in real-world scenarios. As energy demand continues to grow, there is a critical need for more intelligent and efficient prediction systems that can provide precise and timely energy forecasts.

One major research gap in existing energy prediction systems is the limited ability of traditional machine learning models to handle time-dependent and sequential energy consumption data. Although models such as Linear Regression, Decision Trees, and Random Forest can identify important patterns, they often fail to capture long-term temporal dependencies in energy usage. This limitation reduces their effectiveness in forecasting dynamic energy demand, especially in smart buildings and smart grid environments where consumption patterns continuously change over time.

Another important challenge is the lack of model interpretability and transparency in many advanced prediction systems. Deep learning models such as Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) can achieve high prediction accuracy, but they often function as black-box models, making it difficult to understand how predictions are generated. This lack of explainability reduces user trust and limits practical adoption in critical energy management applications. There is a growing need for explainable AI techniques that can provide insights into feature importance and improve decision-making.

Scalability and real-time implementation also remain major research gaps. Many existing studies evaluate their models only on limited datasets or controlled simulation environments, which may not reflect real-world energy systems. Deploying prediction models in large-scale smart city infrastructures requires efficient algorithms capable of handling continuous data streams and adapting to changing conditions. Computational complexity and resource requirements of deep learning models further create challenges for practical implementation.

To address these research gaps, this project focuses on developing an advanced AI-powered energy consumption prediction system that combines multiple machine learning and deep learning models, including hybrid architectures such as Transformer-LSTM. The integration of SHAP explainability further improves transparency by identifying the most influential features affecting energy consumption. By addressing issues

related to prediction accuracy, temporal dependency learning, interpretability, and scalability, the proposed system aims to provide a more reliable and intelligent solution for sustainable energy management in smart environments.

1.4 Constraints

The proposed AI-powered smart energy consumption prediction system is designed by following established software engineering and machine learning standards to ensure reliability, accuracy, and efficiency. A modular system architecture is adopted to separate data preprocessing, feature engineering, model training, prediction, and result visualization into independent components. This design improves system organization, simplifies maintenance, and allows easy integration of new features or predictive models in the future. Standard coding practices and proper documentation are followed to enhance code readability, reproducibility, and long-term maintainability.

Data quality and model development standards are also essential in the system design. The collected energy consumption data is carefully preprocessed through handling missing values, feature scaling, and encoding of categorical variables to ensure consistency and improve model performance. Standard evaluation metrics such as R-squared, Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) are used to measure prediction accuracy and compare model effectiveness. Hyperparameter tuning and validation techniques are applied to optimize model performance and improve generalization on unseen data.

The system is designed to support interoperability and scalability for smart energy environments. It follows compatible data formats and integration standards to enable communication with smart meters, IoT devices, and energy management platforms. Security and privacy standards are incorporated to ensure safe handling of energy-related data and protection against unauthorized access. Additionally, explainable AI techniques such as SHAP are integrated to improve transparency and interpretability, allowing users to understand the key factors influencing energy consumption predictions. These design standards ensure that the system is accurate, efficient, reliable, and suitable for practical deployment in smart buildings and smart city infrastructures.

1.5 Objectives of the Study

The primary objective of this research is to develop an accurate and intelligent AI-powered energy consumption prediction system using machine learning and deep learning techniques for effective energy management in smart environments.

The following are the major objectives of the proposed study:

Development of an AI-Based Energy Consumption Prediction Framework

This objective focuses on designing a complete prediction framework that utilizes his-

torical and real-time energy consumption data to forecast future energy demand in smart buildings and smart grid environments.

Implementation of Machine Learning Models

This involves implementing traditional machine learning algorithms such as Ridge Regression, KNN, Decision Tree, Random Forest, and XGBoost to analyze and predict energy consumption patterns.

Implementation of Deep Learning Models

This objective focuses on applying deep learning techniques such as Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks to capture complex and temporal dependencies in energy usage data.

Hybrid Model Development

This involves developing a hybrid Transformer–LSTM model to enhance learning of both global dependencies and sequential patterns in energy consumption data for improved forecasting capability.

Model Evaluation and Performance Comparison

This objective focuses on evaluating all implemented models using performance metrics such as R-squared, Mean Squared Error (MSE), and Root Mean Squared Error (RMSE), and identifying the best-performing model.

Explainable AI Integration

This involves using SHAP (SHapley Additive exPlanations) to improve interpretability by analyzing the contribution of each feature toward energy consumption predictions.

Deployment and Real-World Applicability

This involves designing the system to be scalable and deployable in real-world smart energy environments such as smart buildings, smart cities, and industrial energy systems.

Sustainable Energy Management Support

This objective focuses on enabling efficient energy usage, reducing wastage, and supporting sustainable energy management through intelligent prediction and optimization techniques.

1.6 Scope of the Project

The scope of this project is to design and develop an AI-powered smart energy consumption prediction system using machine learning, deep learning, and explainable AI techniques. The system aims to accurately forecast energy usage in smart buildings and smart city environments by analyzing historical energy data, environmental conditions,

and temporal features.

Model Architecture

The proposed system implements multiple machine learning models such as Ridge Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and XGBoost for baseline prediction. In addition, deep learning models such as Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks are used for capturing nonlinear and time-dependent patterns. A hybrid Transformer–LSTM architecture is also integrated to improve long-term dependency learning and enhance prediction accuracy.

Dataset Coverage

The project utilizes real-world smart energy datasets containing features such as AC load, lighting load, plug load, temperature, humidity, ambient light, and time-based variables including hour, day, month, and year. These datasets represent energy consumption behavior in smart environments and are used for both training and testing of predictive models.

Learning Approach

The proposed system follows a hybrid learning approach combining machine learning and deep learning techniques. Feature engineering, normalization, and data preprocessing are applied to improve model performance. The system leverages both sequential learning (LSTM) and attention-based learning (Transformer) to capture complex temporal relationships in energy consumption data.

Evaluation Metrics

The performance of the proposed system is evaluated using metrics such as Accuracy, Precision, Recall, F1-score, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and confusion matrix analysis. SHAP (SHapley Additive Explanations) is used to improve model interpretability by identifying the contribution of each feature to the prediction results.

Application Boundary

The proposed system is designed for smart city and smart grid applications where accurate energy forecasting is required for efficient energy management. It can be applied in smart buildings, industrial energy monitoring systems, and urban power distribution networks. However, the current implementation is intended for research and simulation purposes and can be extended for real-time deployment in future work.

1.7 Organization of the Thesis

This thesis is organized into multiple chapters to provide a structured understanding of the proposed system. Chapter 1 introduces the problem statement, motivation, objectives, and research gaps in energy consumption prediction. Chapter 2 presents the literature survey, which reviews existing machine learning, deep learning, and AI-based approaches for energy forecasting and identifies limitations in current methodologies. Chapter 3 describes the proposed methodology, including dataset description, preprocessing techniques, feature engineering, and model architecture such as machine learning models, LSTM, and Transformer-LSTM. Chapter 4 explains the implementation details, experimental setup, and training process of different models used in the system. Chapter 5 presents the results and performance evaluation using different metrics along with SHAP-based explainability analysis. Finally, Chapter 6 concludes the research work and discusses future enhancements such as real-time deployment, edge computing integration, and advanced hybrid AI models for improved energy prediction.

CHAPTER-2

LITERATURE SURVEY

2.1 Literature Review

A literature review provides a systematic analysis of existing research in the field of energy consumption prediction. It helps in understanding previous methodologies, identifying limitations, and discovering research gaps. With the rapid growth of smart cities, IoT devices, and intelligent energy systems, energy forecasting has become an essential research area for efficient energy management and sustainability. Researchers have explored machine learning, deep learning, and artificial intelligence techniques to improve prediction accuracy and system efficiency.

Machine learning techniques have been widely used for energy consumption prediction due to their ability to learn patterns from historical data. Saluja and Palwe [1] proposed ML-based energy prediction models that demonstrated effective forecasting performance. Vasudha et al. [2] analyzed power consumption using smart meters and machine learning methods. Nayak et al. [3] developed state-level forecasting systems using ML algorithms. Raviprabhakaran et al. [4] studied household energy consumption patterns using ML techniques. Abdul Munaf et al. [5] implemented ML models for household energy prediction. Ahmad et al. [6] combined big data analytics with ML for improved forecasting. Alomoush et al. [8] applied ML for residential load prediction in smart cities. These studies show that ML models are effective but limited in capturing temporal dependencies.

Deep learning models have significantly improved energy forecasting by capturing non-linear and sequential patterns. Kushwaha et al. [7] demonstrated that deep learning outperforms traditional ML in urban energy prediction. M. I. A. et al. [9] proposed city-level energy forecasting systems. Helli et al. [10] applied deep learning for smart city energy prediction. Panigrahi et al. [11] used deep learning for demand response systems. Lee et al. [12] introduced edge-based deep learning for real-time prediction. Adhikari et al. [13] reviewed ML-DL hybrid systems and highlighted their advantages. These studies confirm that DL models perform better but require high computational resources.

Artificial intelligence-based approaches have further enhanced smart energy systems by enabling optimization and intelligent decision-making. Wang et al. [14] focused on renewable energy prediction and storage optimization. Campagne et al. [15] used Graph Neural Networks for electricity load forecasting. Mathumitha et al. [16] reviewed deep learning techniques for smart buildings. Sharma et al. [17] introduced causal neural networks for energy forecasting. Kolluru et al. [18] developed LSTM-based AI energy optimization systems integrated with IoT data. These studies highlight the importance of intelligent, adaptive, and automated energy systems.

Recent research focuses on explainable AI and hybrid models for improving accuracy and interpretability. Munir et al. [19] used LightGBM with explainable AI techniques

for energy prediction. Dağkurs and Atacak [20] proposed hybrid neuro-fuzzy deep learning models. Hamayat et al. [21] developed deep learning-based prediction and anomaly detection systems. Kumar et al. [22] introduced hybrid deep learning models for smart grid forecasting. Chen et al. [23] proposed Transformer-based energy forecasting models. Patel et al. [24] applied SHAP-based explainable AI for energy prediction. Lee et al. [25] developed LSTM-attention hybrid models for smart city forecasting. These studies show that hybrid and explainable AI models improve both accuracy and transparency.

Overall, the literature indicates that ML models are simple and interpretable, DL models provide higher accuracy, and hybrid AI models aim to balance both. However, challenges such as interpretability, computational cost, and dataset dependency still exist. Therefore, this study focuses on developing an LSTM-based energy prediction system with explainable AI to achieve accurate and interpretable forecasting.

2.2 Research Gap and Contributions

Research Gaps :

- Existing traditional machine learning models such as Linear Regression, Decision Tree, and Support Vector Regression (SVR) are limited in capturing complex nonlinear and temporal relationships in energy consumption data.
- Many deep learning models improve accuracy but require high computational resources, making them less suitable for real-time deployment in resource-constrained environments.
- Most existing studies focus only on either machine learning or deep learning models, with limited comparison between multiple model categories in a unified framework.
- Hybrid architectures such as Transformer–LSTM are still under-explored in many energy prediction applications and often lack consistent performance across different datasets.
- A major limitation in existing systems is the lack of interpretability, making it difficult to understand how predictions are generated and which features influence them.
- Many existing approaches do not effectively integrate explainable AI techniques to provide transparent decision-making in energy forecasting systems.
- Limited studies focus on combining environmental, temporal, and load-based features in a single unified prediction framework for smart energy systems.

Contributions of the Proposed Work :

- Development of an intelligent energy consumption prediction system using multiple machine learning models for baseline comparison.
- Implementation of deep learning models, particularly Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM), for capturing complex nonlinear and time-dependent patterns in energy data.
- Design and evaluation of a hybrid deep learning approach to improve prediction accuracy and robustness.
- Comprehensive comparison of machine learning and deep learning models to identify the most effective approach for energy forecasting.
- Integration of feature engineering techniques including normalization, handling missing values, and temporal feature extraction to improve model performance.
- Application of SHAP-based explainable AI to enhance transparency by identifying the contribution of each feature in predictions.
- Development of a scalable and efficient framework suitable for smart city and smart grid energy management applications.
- Improvement of prediction accuracy and generalization capability through systematic model tuning and evaluation.

CHAPTER-3

PROPOSED SYSTEM

The AI-Powered Smart Energy Consumption Prediction System begins with collecting raw energy and environmental sensor data from multiple smart building datasets. The collected data undergoes preprocessing techniques such as handling missing values using mean imputation, converting date columns into datetime format, and extracting temporal attributes like year, month, day, hour, and minute. Feature scaling is then applied using Min-Max normalization to standardize numerical values between 0 and 1. Finally, the dataset is divided into training and testing sets for model training and evaluation.

The preprocessed dataset is provided to various Machine Learning and Deep Learning models including Ridge Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, XGBoost, Artificial Neural Network (ANN), Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM). The overall architecture of the proposed smart energy forecasting system is illustrated in Figure 3.1.

The system predicts future energy consumption based on environmental conditions such as temperature, humidity, lighting load, plug load, ambient light, and peak usage hours. SHAP analysis is used to explain the importance of features influencing prediction results. Finally, the system evaluates model performance using metrics such as Accuracy, Precision, Recall, and F1-Score. The final output of the system is an accurate prediction of smart building energy consumption, helping reduce power wastage and optimize energy management.

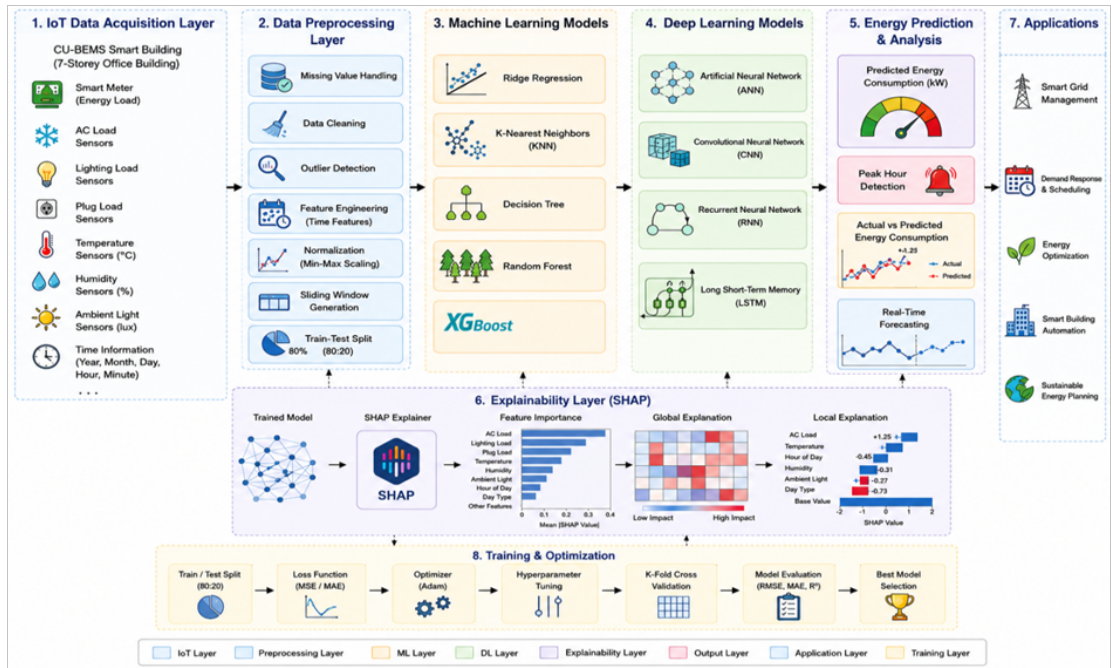


Figure 3.1: Architecture of the proposed system

3.1 Input Dataset

The first step in the proposed system involves collecting smart building energy consumption data from multiple CSV files. The dataset contains historical information related to electricity usage and environmental conditions inside smart buildings. A total of 14 CSV files are merged into a single consolidated dataset for analysis and prediction purposes.

The dataset contains several important features as shown in Figure 3.2 are associated with energy consumption behavior. These features help the prediction models understand how environmental and operational conditions affect electricity usage patterns in smart buildings.

...	Date	z1_Light(kW)	z1_Plug(kW)	z2_AC1(kW)	z2_AC2(kW)	z2_AC3(kW)	z2_AC4(kW)	z2_Light(kW)	z2_Plug(kW)	z3_Light(kW)	...
0	2018-07-01 00:00:00	12.94	18.56	45.24	0.01	0.01	0.00	13.76	17.64	10.92	...
1	2018-07-01 00:01:00	12.97	18.55	45.28	0.02	0.01	0.01	13.76	17.21	10.95	...
2	2018-07-01 00:02:00	12.97	18.55	45.24	0.01	0.01	0.01	13.79	17.18	10.94	...
3	2018-07-01 00:03:00	12.98	18.58	45.26	0.02	0.01	0.00	13.81	16.64	10.94	...
4	2018-07-01 00:04:00	13.01	18.60	45.22	0.02	0.01	0.01	13.83	15.69	10.97	...

5 rows × 49 columns

Figure 3.2: Various features in the dataset

3.1.1 Detailed Features of the Dataset

Temperature (Numeric - Float)

This feature represents the temperature inside the smart building environment. Temperature significantly affects energy consumption because cooling and heating appliances consume more power during extreme conditions.

Humidity (Numeric - Float)

The humidity feature indicates the moisture content in the environment. Variations in humidity influence HVAC system usage and overall energy demand.

Ambient Light (Numeric - Float)

This feature measures the natural light intensity available in the building. Ambient light affects lighting usage and helps optimize electricity consumption.

Lighting Load (Numeric - Float)

The lighting load feature represents the energy consumed by lighting systems in the building. Higher lighting requirements increase overall electricity usage.

Plug Load (Numeric - Float)

Plug load refers to the energy consumed by connected electrical devices such as computers, chargers, and appliances. It contributes significantly to total energy demand.

AC Load (Numeric - Float)

This feature indicates the electricity consumed by air conditioning systems. AC load is one of the major contributors to building energy consumption.

Date and Time (Date-Time Object)

The date and time features record the exact timestamp for each energy reading. These attributes help analyze energy usage trends across different periods and enable time-series forecasting.

Peak Hour (Categorical)

The Peak Hour feature identifies periods of high energy demand. It helps classify energy usage into high and low consumption categories.

Total Load (Numeric - Float)

The Total Load feature represents the total electricity consumed in the smart building. It acts as the target variable for prediction and forecasting models.

3.2 Data Preprocessing

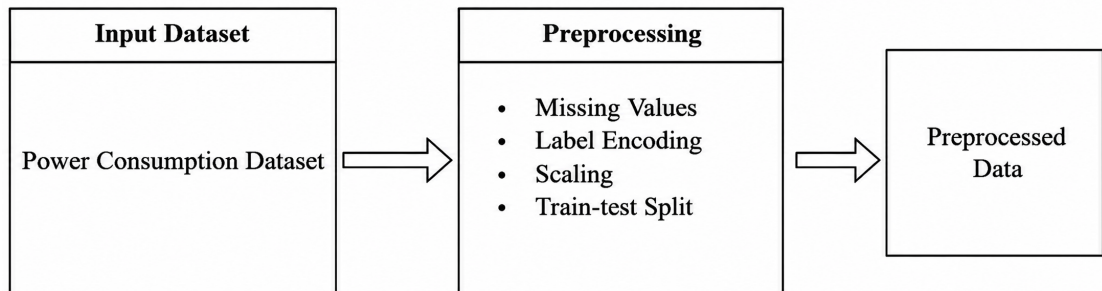


Figure 3.3: Pre-processing Overview

Figure 3.3 illustrates the pre-processing overview of the proposed AI-Powered Smart Energy Consumption Prediction System. Data pre-processing is one of the most important stages in the system, as it prepares raw smart building energy data into a clean, structured, and machine-readable format suitable for Machine Learning and Deep Learning models. The pre-processing stage improves data quality, removes inconsistencies, and enhances prediction accuracy.

The collected smart building dataset contains environmental parameters, appliance load values, and time-based information. Before model training, the dataset undergoes multiple preprocessing operations including missing value handling, feature engineering, normalization, and train-test splitting. These operations ensure reliable forecasting and efficient model performance.

Through preprocessing, the system becomes capable of identifying meaningful energy consumption patterns, reducing noise in the data, and improving the accuracy of prediction models such as Random Forest, XGBoost, ANN, RNN, and LSTM. The overall objective is to prepare optimized input data for accurate smart energy forecasting and analysis.

3.2.1 Missing values

The preprocessing stage begins by checking the dataset for missing or incomplete values. Since the smart building dataset is collected from multiple IoT sensors and CSV files, some records may contain missing entries due to sensor failures, transmission delays, or inconsistent measurements. Missing values can negatively affect model performance if not handled properly.

The dataset includes numerical features such as temperature, humidity, lighting load, plug load, and AC load. Missing numerical values are handled using mean imputation, where the missing entries are replaced with the average value of the respective feature column. This helps maintain data consistency without removing useful records from the dataset.

Handling missing values is essential because incomplete environmental readings can reduce prediction accuracy and create inconsistencies in energy forecasting models. By performing data imputation, the dataset becomes more reliable for training machine learning algorithms

Table 3.1 shows the missing values present in the smart building energy dataset before preprocessing. The analysis indicates that several sensor and appliance load features contain large numbers of missing entries, requiring preprocessing and data cleaning.

Parameters used for missing value handling

Mean Imputation

Mean imputation is used to replace missing numerical values with the average value of the corresponding feature. This method helps preserve dataset size and reduces information loss.

Data Cleaning

Data cleaning removes duplicate records, inconsistent entries, and unwanted noise from the dataset to improve data quality.

Date-Time Conversion

The Date column is converted into datetime format to enable time-series analysis and extraction of temporal features.

Temporal Feature Extraction

Additional features such as Year, Month, Day, Hour, and Minute are extracted from the

Table 3.1: Missing Values Count Before Preprocessing

Feature	Missing Values Count
Date	0
z1_Light(kW)	158487
z1_Plug(kW)	105621
z2_AC1(kW)	244978
z2_AC2(kW)	3856173
z2_AC3(kW)	3856174
z2_AC4(kW)	3856174
z2_Light(kW)	180556
z2_Plug(kW)	208328
z3_Light(kW)	108142
z3_Plug(kW)	109015
z4_Light(kW)	131433
z1_AC1(kW)	1079995
z1_S1(degC)	2489141
z1_S1(RH%)	2489139
z1_S1(lux)	2489202
z2_AC5(kW)	4643072
z2_AC6(kW)	4643072
z2_AC7(kW)	4643072
z2_AC8(kW)	4643072
z2_AC9(kW)	4643072
z2_AC10(kW)	4643072
z2_AC11(kW)	4643072
TOTAL Missing Values	135855677

datetime column to capture energy consumption trends over time.

Feature Aggregation

Individual appliance loads are aggregated into higher-level categories such as: • AC Load • Lighting Load • Plug Load This simplifies energy usage analysis and improves forecasting performance.

Peak Hour Feature

A Peak Hour feature is created to identify high energy demand intervals. This helps classify energy usage into high and low consumption periods.

Min-Max Normalization

Numerical features are scaled between 0 and 1 using Min-Max normalization. Feature scaling ensures equal contribution of all parameters during model training.

Train-Test Split

The dataset is divided into: • Training Set (80%) • Testing Set (20%) This allows the models to learn from training data and evaluate performance using unseen testing data.

3.2.2 Date-Time Conversion

The Date column is converted into datetime format to enable time-series analysis and extraction of temporal features.

Converting Date Column

The Date column is transformed into a machine-readable datetime format for efficient processing and analysis of smart building energy consumption patterns.

Extracting Year, Month, Day, Hour, Minute

Additional temporal features such as Year, Month, Day, Hour, and Minute are extracted from the datetime column to capture energy consumption trends over time.

3.2.3 Data Encoding

Data encoding is the process of converting categorical information into numerical format so that Machine Learning and Deep Learning models can process the data efficiently.

Converting Categorical Values - In the proposed system, categorical features such as Peak Hour categories and other classification labels are transformed into machine-readable numerical values such as 0 and 1. Data encoding improves model compatibility and enables efficient learning during training and prediction processes.

Peak Hour Encoding-

The Peak Hour feature is encoded to identify periods of high electricity demand and classify energy consumption into High Energy Usage and Low Energy Usage categories based on operational conditions and peak demand intervals.

3.2.4 Feature Scaling

Feature scaling is performed to ensure balanced learning among all input features in the dataset.

Min-Max Normalization- In the proposed system, Min-Max normalization is applied to scale numerical values into a standardized range between 0 and 1. Feature scaling improves model convergence speed, enhances training stability, and increases overall prediction accuracy.

3.2.5 Dataset Splitting

After preprocessing and normalization, the dataset is divided into training and testing datasets for model development and evaluation.

The training dataset is used for learning energy consumption patterns and training Machine Learning and Deep Learning models such as ANN, CNN, RNN, and LSTM. The testing dataset is used for evaluating prediction performance and analyzing forecasting accuracy using unseen smart building energy consumption data.

3.3 Feature Engineering

Feature engineering is performed to improve prediction efficiency and simplify energy analysis. The proposed system creates meaningful input features from raw smart building energy data to improve Machine Learning and Deep Learning model performance.

AC Load Aggregation: Individual air conditioning energy consumption values are aggregated into a single AC Load feature. This feature represents the total electricity consumed by air conditioning systems and helps analyze cooling-related energy demand inside smart buildings.

Lighting Load Aggregation: Lighting energy consumption values are combined into a single Lighting Load feature. This feature represents the total electricity consumed by lighting systems and helps identify lighting-related energy usage patterns.

Plug Load Aggregation: Energy consumption values from electrical appliances and connected devices are aggregated into a single Plug Load feature. This feature includes electricity usage from computers, chargers, office equipment, and other electronic appliances.

Peak Hour Creation: A Peak_Hour feature is created to identify periods of high electricity demand inside smart buildings. This feature helps classify energy usage into high and low consumption categories and supports intelligent energy management.

Temporal Feature Extraction: Additional temporal features such as Year, Month, Day, Hour, and Minute are extracted from the datetime column to capture time-dependent energy consumption patterns and improve forecasting accuracy.

3.4 Model Building

In the proposed AI-Powered Smart Energy Consumption Prediction System, various Machine Learning and Deep Learning models are used to analyze smart building energy consumption patterns and predict future electricity demand. The prediction models are trained using environmental parameters, appliance load information, and temporal features extracted during preprocessing and feature engineering stages. The system takes important input features such as temperature, humidity, ambient light, AC load, plug load, lighting load, Peak Hour, and time-related information to forecast future energy usage accurately.

Machine Learning Models

The proposed system utilizes Machine Learning algorithms such as Ridge Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, and XGBoost to analyze relationships between environmental conditions and electricity consumption patterns. These algorithms help forecast future energy demand and improve prediction performance using historical smart building energy data.

Deep Learning Models

The proposed system also utilizes Deep Learning models such as Artificial Neural Network (ANN), Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM) to learn complex nonlinear and sequential energy consumption patterns from time-series smart building data. Among these models, LSTM plays a major role because of its ability to preserve long-term temporal dependencies and generate highly accurate smart building energy forecasts.

3.4.1 LSTM Algorithm

The proposed AI-Powered Smart Energy Consumption Prediction System utilizes several Machine Learning and Deep Learning models for accurate energy forecasting. Among these models, the Long Short-Term Memory (LSTM) network is mainly used for predicting future smart building energy consumption because of its ability to learn sequential and time-dependent patterns from historical data.

The LSTM algorithm processes environmental and energy-related features such as temperature, humidity, ambient light, AC load, lighting load, plug load, peak hour, and time-based information to forecast future energy usage. The algorithm works through multiple stages including initialization, forward propagation, memory cell processing, error calculation, backpropagation, and optimization.

1. Initialize Model Parameters

Initially, the weights and biases of the LSTM model are initialized with small random values. These parameters are adjusted during training to improve prediction accuracy. The training process begins by feeding smart building energy data into the network.

The dataset includes important features such as:

- Temperature
- Humidity
- Ambient Light

- AC Load
- Plug Load
- Lighting Load
- Peak Hour
- Date and Time

These parameters are used as input variables for energy forecasting.

2. Forward Propagation

During forward propagation, the input data passes through the LSTM network. The LSTM model contains specialized memory cells that help preserve long-term sequential information related to energy consumption patterns.

The hidden state of the LSTM is calculated as:

$$h_t = o_t \cdot \tanh(c_t) \quad (3.1)$$

The LSTM network uses gates to regulate information flow:

- Input Gate
- Forget Gate
- Output Gate

These gates help the model retain useful information and discard irrelevant data during prediction.

3. Error Calculation

After generating predictions, the system compares predicted energy consumption values with actual values from the testing dataset. The prediction error is computed using the Mean Squared Error (MSE) loss function.

The Mean Squared Error is represented as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3.2)$$

The main objective of the model is to minimize prediction error and improve forecasting performance.

4. Backpropagation and Optimization

The error calculated during prediction is propagated backward through the network using the backpropagation algorithm. During this process, the weights and biases are updated to improve model accuracy.

The weight update equation is:

$$W_{new} = W_{old} - \eta \frac{\partial L}{\partial W} \quad (3.3)$$

where:

- W_{new} = Updated Weight
- W_{old} = Previous Weight
- η = Learning Rate
- $\frac{\partial L}{\partial W}$ = Gradient of Loss Function

Adam Optimizer- The Adam optimizer is used to improve convergence speed, optimize training performance, and reduce prediction error during model training.

5. Energy Usage Classification

After predicting future energy consumption, the system classifies energy usage into:

- High Energy Usage represents periods of increased electricity demand caused by higher appliance usage and operational activity inside smart buildings.
- Low Energy Usage represents periods of reduced electricity demand and lower operational energy consumption.

The Peak_Hour feature helps identify periods of high electricity demand and supports efficient energy management.

6. SHAP Analysis

Finally, SHAP (SHapley Additive Explanations) analysis is used to identify the most influential features affecting prediction results.

Important features influencing energy prediction include:

- AC Load
- Lighting Load
- Plug Load
- Temperature

- Ambient Light

Explainability Analysis helps understand how individual input features contribute to prediction results and supports intelligent smart energy management decisions.

3.4.2 Transformer + LSTM Model

In addition to the LSTM model, the proposed system utilizes a hybrid Transformer-LSTM architecture to improve smart building energy forecasting performance. The Transformer model uses a self-attention mechanism to identify important temporal patterns and relationships from time-series energy consumption data.

The Transformer component extracts important feature representations from parameters such as temperature, humidity, ambient light, AC load, plug load, lighting load, and Peak Hour information. These extracted sequential features are then provided to the LSTM network for learning long-term temporal dependencies and generating accurate future energy consumption predictions.

The hybrid Transformer-LSTM model combines the advantages of both architectures. The Transformer improves feature extraction and contextual learning, while the LSTM preserves sequential memory information. This combination improves forecasting accuracy, reduces prediction error, and enhances the reliability of the proposed smart energy prediction system.

3.5 Model Evaluation

Model evaluation is one of the most important stages in the AI-Powered Smart Energy Consumption Prediction System. It involves analyzing the performance, accuracy, reliability, and prediction capability of Machine Learning and Deep Learning models using unseen testing data. The evaluation process helps determine how effectively the trained models can predict future smart building energy consumption under real-world conditions.

In the proposed system, multiple Machine Learning and Deep Learning algorithms such as Ridge Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, XGBoost, Artificial Neural Network (ANN), Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), and Long Short-Term Memory (LSTM) are evaluated and compared using different performance metrics. The evaluation stage helps identify the most accurate and efficient model for energy forecasting and smart energy management applications.

Importance of Model Evaluation

Quality Assurance: Model evaluation ensures that the prediction model can accurately forecast future energy consumption when exposed to new and unseen smart building data. It validates the model's ability to generalize energy consumption patterns effectively and improves system reliability.

Comparing Models: The proposed system uses multiple Machine Learning and Deep Learning algorithms. Model evaluation helps compare these algorithms based on prediction performance and enables selection of the most suitable forecasting model for smart energy management.

Performance Optimization: Evaluation metrics reveal areas where the model performs poorly. This information helps improve training strategies, optimize hyperparameters, reduce prediction errors, and enhance overall model accuracy.

Decision Support: Prediction performance directly influences intelligent energy management decisions such as electricity optimization, peak load management, and energy conservation planning. A properly evaluated model provides confidence for practical deployment in smart building environments.

Real-Time Deployment: A thoroughly evaluated model is more reliable for real-world implementation. High-performing prediction models can be integrated into smart building systems for real-time monitoring and energy forecasting applications.

Performance Metrics:

The proposed system evaluates model performance using multiple classification and regression metrics. These evaluation metrics help measure prediction accuracy, classification capability, and forecasting efficiency.

Accuracy: It measures the overall correctness of the prediction model by calculating the proportion of correctly predicted energy usage instances compared to the total number of predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.4)$$

where:

- TP = True Positives
- TN = True Negatives

- FP = False Positives
- FN = False Negatives

Precision: It measures how many predicted positive energy usage instances are actually correct. It evaluates the reliability of positive predictions generated by the model.

$$Precision = \frac{TP}{TP + FP} \quad (3.5)$$

Recall: It measures the ability of the model to correctly identify actual positive energy usage instances. It indicates how effectively the model detects high energy consumption patterns.

$$Recall = \frac{TP}{TP + FN} \quad (3.6)$$

F1-Score: It is the harmonic mean of Precision and Recall. It provides a balanced evaluation metric when dealing with classification tasks involving high and low energy usage categories.

$$F1-Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (3.7)$$

Mean Absolute Error (MAE): This gives the average absolute difference between actual and predicted energy values. Lower MAE values represent more accurate predictions.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3.8)$$

Mean Squared Error (MSE): It is used to evaluate regression-based energy forecasting models. It measures the average squared difference between actual energy values and predicted energy values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3.9)$$

Root Mean Squared Error (RMSE): This is the square root of the Mean Squared Error. RMSE represents prediction error in the same unit as the target variable, making interpretation easier.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.10)$$

R-Squared Score (R^2): It also known as the coefficient of determination, measures how well the independent variables explain variations in energy consumption. Higher R^2 values indicate better model fitting and stronger prediction capability.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (3.11)$$

Confusion Matrix

The confusion matrix is used to evaluate classification performance for High and Low energy usage prediction. It provides information about correctly and incorrectly classified instances, including True Positives, True Negatives, False Positives, and False Negatives. The confusion matrix helps analyze the strengths and weaknesses of the prediction models and supports performance improvement.

Result Analysis

The proposed system compares the performance of Machine Learning and Deep Learning models using various evaluation metrics. Experimental analysis shows that Deep Learning models such as LSTM achieve better prediction accuracy because of their ability to learn sequential energy consumption patterns from time-series smart building data. The evaluation results demonstrate that the proposed system provides accurate and reliable smart energy forecasting for intelligent energy management applications.

3.6 Constraints

Several constraints influence the design and implementation of the proposed AI-Powered Smart Energy Consumption Prediction System.

1. Data Availability

The prediction system depends on historical smart building energy datasets collected from IoT sensors and CSV files. Limited or incomplete datasets can affect model learning capability and prediction accuracy.

2. Data Quality

Sensor-generated data may contain missing values, noisy readings, duplicate records, and inconsistent measurements. Poor-quality data can negatively affect Machine Learning and Deep Learning model performance.

3. Computational Complexity

Deep Learning models such as ANN, CNN, RNN, and LSTM require high computational resources, memory usage, and processing time during model training and prediction.

4. Real-Time Deployment

Implementing real-time energy forecasting systems requires continuous data acquisition, preprocessing, and prediction. Hardware limitations and communication delays may affect real-time monitoring performance.

5. Resource Availability

The proposed system requires sufficient hardware resources, storage capacity, and software frameworks such as TensorFlow, Keras, and Scikit-learn for efficient model training and deployment.

6. Model Generalization

Energy consumption patterns may vary across different smart buildings and environments. Models trained on one dataset may require retraining or fine-tuning for effective performance in other real-world scenarios.

3.7 Cost and Sustainability Impact

The AI-Powered Smart Energy Consumption Prediction System contributes significantly to energy optimization, operational efficiency, economic savings, and environmental sustainability. The proposed system supports intelligent decision-making for smart building management and promotes sustainable energy utilization.

1. Cost Efficiency

The proposed system improves cost efficiency by accurately forecasting future electricity demand and identifying high energy consumption periods. Smart energy prediction helps reduce unnecessary power wastage, optimize appliance usage, and minimize electricity bills for smart building operators and users.

2. Energy Optimization

By predicting future energy demand, the system enables efficient scheduling and operation of electrical appliances such as air conditioning systems, lighting systems, and plug loads. This reduces excessive power usage and improves overall energy efficiency inside smart buildings.

3. Environmental Sustainability

The proposed system indirectly supports environmental sustainability by reducing unnecessary electricity consumption and minimizing carbon emissions associated with power generation. Optimized energy usage contributes to cleaner and greener smart building environments.

4. Resource Efficiency

Intelligent energy forecasting helps optimize the utilization of electrical infrastructure and energy resources. Efficient energy management reduces overload conditions, minimizes power losses, and improves operational efficiency in smart buildings.

5. Economic Benefits

Reducing energy wastage and optimizing electricity usage leads to long-term economic benefits for residential buildings, commercial organizations, and industries. Lower operational costs improve financial efficiency and support sustainable smart building operations.

6. Smart Building Automation

The prediction system supports intelligent automation by enabling proactive energy management decisions. Automated energy control systems can adjust appliance operations based on predicted consumption levels, improving user comfort and operational efficiency.

7. Peak Load Management

The Peak Hour feature helps identify periods of high electricity demand. Effective peak load management reduces stress on electrical infrastructure and prevents unnecessary energy overload conditions.

8. Improved Decision Making

The integration of SHAP explainability analysis improves transparency by identifying the most influential factors affecting energy consumption prediction. This helps building administrators and energy managers make informed decisions regarding electricity optimization and energy planning.

9. Sustainable Smart Building Development

The proposed system contributes to the development of sustainable smart buildings by integrating AI-driven forecasting techniques with intelligent energy management strategies. It promotes environmentally responsible energy usage and supports future smart city initiatives.

10. Quality of Life Improvement

Efficient energy management improves indoor environmental conditions, appliance performance, and operational comfort inside smart buildings. Reduced electricity wastage and optimized energy utilization contribute to better living and working environments for users.

3.8 System Setup and Requirements

The proposed AI-Powered Smart Energy Consumption Prediction System is implemented using the Python programming language along with Machine Learning and Deep Learning libraries such as TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, and SHAP. These libraries are used for data preprocessing, visualization, model training, performance evaluation, and explainability analysis.

The development and experimentation of the proposed system are carried out in a GPU-enabled environment to improve computational efficiency and accelerate Deep Learning model training. The system is designed to support scalable smart building energy forecasting and can be extended for real-world smart energy management applications.

Hardware Requirements

- Processor: Intel Core i5/i7 or equivalent
- RAM: Minimum 8 GB
- GPU: NVIDIA GPU recommended for faster Deep Learning training
- Storage: Minimum 20 GB free space

Software Requirements

- Operating System: Windows/Linux
- Python Version: Python 3.x
- TensorFlow and Keras
- Scikit-learn
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SHAP Library
- Jupyter Notebook / Google Colab

The proposed system utilizes Deep Learning models such as ANN, CNN, RNN, and LSTM for intelligent smart building energy forecasting. GPU acceleration helps reduce

training time and improves model optimization efficiency during large-scale energy data processing.

The framework is scalable, computationally efficient, and suitable for deployment in intelligent smart building environments for real-time energy monitoring, forecasting, and optimization applications.

CHAPTER-4

IMPLEMENTATION

4.1 Sample Code

```
import numpy as np
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score, precision_score,
    recall_score, f1_score, confusion_matrix

from sklearn.ensemble import RandomForestRegressor
from sklearn.neighbors import KNeighborsRegressor
from sklearn.linear_model import Ridge
from sklearn.tree import DecisionTreeRegressor
from xgboost import XGBRegressor

import tensorflow as tf
from tensorflow.keras.models import Sequential, Model
from tensorflow.keras.layers import Dense, LSTM, GRU, Conv1D,
    Flatten, Input, Dropout

import shap

# Load CU-BEMS dataset (Kaggle)
data = pd.read_csv("CU-BEMS.csv")

# Preprocessing
data.fillna(method='ffill', inplace=True)

# Feature selection
X = data.drop(["energy_consumption"], axis=1)
y = data["energy_consumption"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```

# ----- ML MODELS ----- #

rf = RandomForestRegressor()
knn = KNeighborsRegressor()
ridge = Ridge()
dt = DecisionTreeRegressor()
xgb = XGBRegressor()

rf.fit(X_train, y_train)
knn.fit(X_train, y_train)
ridge.fit(X_train, y_train)
dt.fit(X_train, y_train)
xgb.fit(X_train, y_train)

# ----- DL MODELS ----- #

# ANN
ann = Sequential([
    Dense(64, activation='relu',
input_shape=(X_train.shape[1],)),
    Dense(32, activation='relu'),
    Dense(1)
])

ann.compile(optimizer='adam', loss='mse')
ann.fit(X_train, y_train, epochs=20, batch_size=32)

# CNN (1D)
cnn = Sequential([
    Conv1D(64, kernel_size=2, activation='relu',
input_shape=(X_train.shape[1], 1)),
    Flatten(),
    Dense(32, activation='relu'),
    Dense(1)
])

# reshape for CNN
X_train_cnn = X_train.reshape(X_train.shape[0],
    X_train.shape[1], 1)

```

```

X_test_cnn = X_test.reshape(X_test.shape[0], X_test.shape[1],
                             1)

cnn.compile(optimizer='adam', loss='mse')
cnn.fit(X_train_cnn, y_train, epochs=20, batch_size=32)

# RNN / LSTM
lstm = Sequential([
    LSTM(64, return_sequences=True,
         input_shape=(X_train.shape[1], 1)),
    LSTM(32),
    Dense(1)
])

lstm.compile(optimizer='adam', loss='mse')
lstm.fit(X_train_cnn, y_train, epochs=20, batch_size=32)

# Hybrid TransformerLSTM (simplified)
inputs = Input(shape=(X_train.shape[1], 1))
x = LSTM(64, return_sequences=True)(inputs)
x = LSTM(32)(x)
outputs = Dense(1)(x)

hybrid_model = Model(inputs, outputs)
hybrid_model.compile(optimizer='adam', loss='mse')
hybrid_model.fit(X_train_cnn, y_train, epochs=20,
                 batch_size=32)

# ----- PREDICTION ----- #

y_pred = lstm.predict(X_test_cnn)

# Convert Regression Classification (Peak / Non-Peak)
threshold = y_train.mean()

y_test_class = (y_test > threshold).astype(int)
y_pred_class = (y_pred.flatten() > threshold).astype(int)

# Metrics
acc = accuracy_score(y_test_class, y_pred_class)
prec = precision_score(y_test_class, y_pred_class)

```

```

rec = recall_score(y_test_class, y_pred_class)
f1 = f1_score(y_test_class, y_pred_class)

print("Accuracy:", acc)
print("Precision:", prec)
print("Recall:", rec)
print("F1 Score:", f1)

# ----- SHAP EXPLAINABILITY ----- #

explainer = shap.Explainer(rf)
shap_values = explainer(X_test)

shap.summary_plot(shap_values, X_test)

# ----- SAMPLE OUTPUT ----- #

results = pd.DataFrame({
    "Actual Energy": y_test[:10].values,
    "Predicted Energy": y_pred.flatten()[:10],
    "Actual Peak/Non-Peak": y_test_class[:10].values,
    "Predicted Peak/Non-Peak": y_pred_class[:10]
})

print(results)

```

4.2 Tools and Technologies

Programming Language

- Python 3.10

Python is used for implementing both machine learning and deep learning models due to its simplicity and rich ecosystem.

Machine Learning Models

- Random Forest
- K-Nearest Neighbors (KNN)
- Ridge Regression

- Decision Tree
- XGBoost

These models are used for baseline energy consumption prediction and comparative analysis.

Deep Learning Models

- Artificial Neural Network (ANN)
- Convolutional Neural Network (CNN)
- Recurrent Neural Network (RNN)
- Long Short-Term Memory (LSTM)
- Hybrid Transformer–LSTM

These models are used to capture nonlinear and temporal dependencies in energy consumption data.

Explainability Technique

- SHAP (SHapley Additive Explanations)

SHAP is used to interpret model predictions and identify important features influencing energy usage.

Evaluation Approach

- MSE, MAE, RMSE, R2-Score
- Regression to Classification Conversion (Peak Hour / Non-Peak Hour)
- Accuracy, Precision, Recall, F1-score
- Confusion Matrix Analysis
- ROC Curve

The system is evaluated on the CU-BEMS dataset (Kaggle), and both predicted and actual values are compared for energy consumption as well as peak-hour classification to validate model performance.

4.3 Implementation Summary

The proposed AI-powered energy consumption prediction system integrates both machine learning and deep learning approaches. Machine learning models such as Random Forest, KNN, Ridge Regression, Decision Tree, and XGBoost are used for baseline prediction, while deep learning models including ANN, CNN, RNN, LSTM, and Hybrid Transformer–LSTM are used for advanced temporal modeling.

The regression outputs are converted into classification labels to identify peak and non-peak energy consumption hours. The system is evaluated using Regression metrics such as MSE, MAE, RMSE, R2-Score and classification metrics such as accuracy, precision, recall, and F1-score. SHAP analysis is applied to improve interpretability and understand feature contributions.

Finally, the model is tested on unseen data, and results are displayed by comparing actual vs predicted energy consumption along with actual vs predicted peak-hour classification, demonstrating the effectiveness of the proposed framework.

CHAPTER-5

EXPERIMENTAL RESULTS AND DISCUSSION

5.1 Experimentation and Result Analysis

The experimentation and result analysis phase is one of the most significant stages in the proposed AI-Powered Smart Energy Consumption Prediction System. This section focuses on evaluating the behavior of the smart building energy dataset, understanding feature relationships, analyzing energy consumption patterns, and comparing the performance of various Machine Learning and Deep Learning models used for forecasting future energy demand.

The proposed system utilizes historical smart building energy consumption data along with environmental parameters such as temperature, humidity, ambient light, AC load, lighting load, plug load, and time-based features. Through experimental analysis, the system identifies hidden energy consumption patterns and evaluates the forecasting capability of the developed prediction models.

Various graphical visualization techniques and statistical performance metrics are used to interpret the dataset and evaluate the effectiveness of the proposed prediction system.

5.1.1 Energy Consumption Across Different Environmental Conditions

To analyze the impact of environmental conditions on smart building energy consumption, the proposed system uses box plot visualization as shown in Figure 6. The box plot represents statistical measures such as median, quartiles, interquartile range (IQR), minimum and maximum values, and outliers to study variations in electricity usage under different conditions.

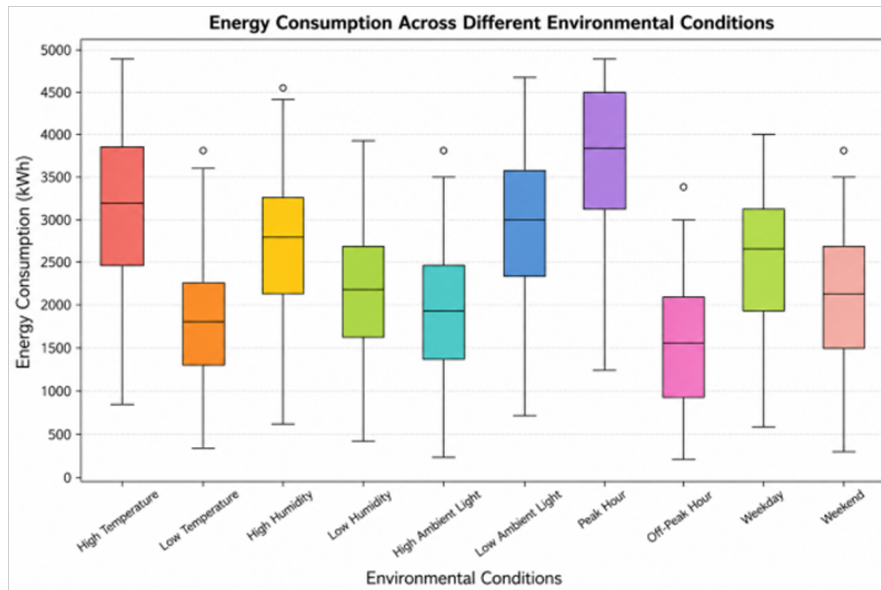


Figure 5.1: Energy Consumption Across Various Environmental Conditions

Figure 5.1 shows that energy consumption is higher during high-temperature and peak-

hour conditions due to increased AC and appliance usage. Weekdays also exhibit higher electricity demand compared to weekends because of greater operational activities and occupancy levels. In contrast, increased ambient light reduces dependency on artificial lighting, resulting in lower energy consumption.

The analysis also identifies abnormal energy usage patterns and outliers, which help improve intelligent energy monitoring and optimization in smart building environments.

5.1.2 Energy Consumption by Time of Day

Time-series analysis is an important component of the proposed system because smart building energy consumption varies throughout the day based on occupancy, operational activities, and environmental conditions. Figure 5.2 illustrates the variation of energy consumption across different hours of the day.

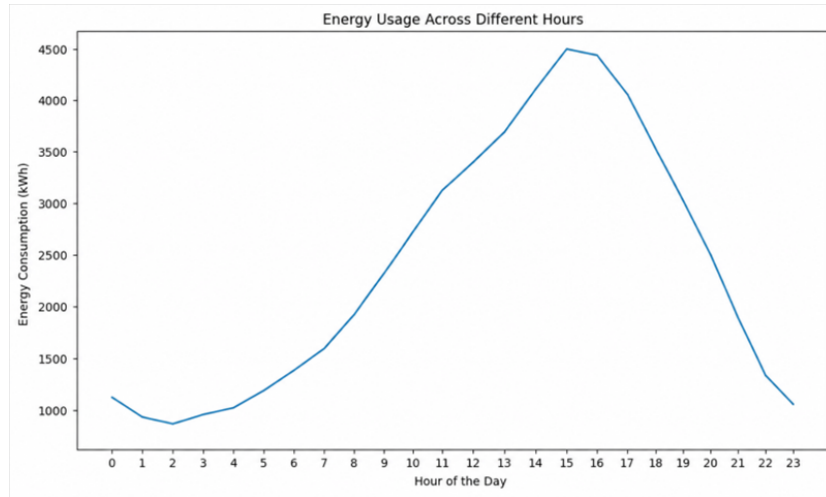


Figure 5.2: Energy Usage Across Different Hours

The graph shows that electricity consumption gradually increases during morning hours as appliances and systems become operational, and reaches its peak during afternoon and evening hours due to higher occupancy and appliance usage. The `Peak_Hour` feature effectively identifies these high-demand intervals.

Energy consumption decreases during late-night and early-morning hours because of reduced occupancy and minimal appliance usage. This time-based analysis improves the performance of sequential Deep Learning models such as RNN and LSTM by helping them learn temporal energy consumption patterns.

5.1.3 Correlation Matrix Analysis

The proposed system uses a correlation matrix to analyze the relationships between environmental parameters, appliance loads, and total energy consumption. Figure 5.3

presents a color-coded heatmap where positive correlations indicate direct relationships, negative correlations indicate inverse relationships, and darker colors represent stronger correlations.

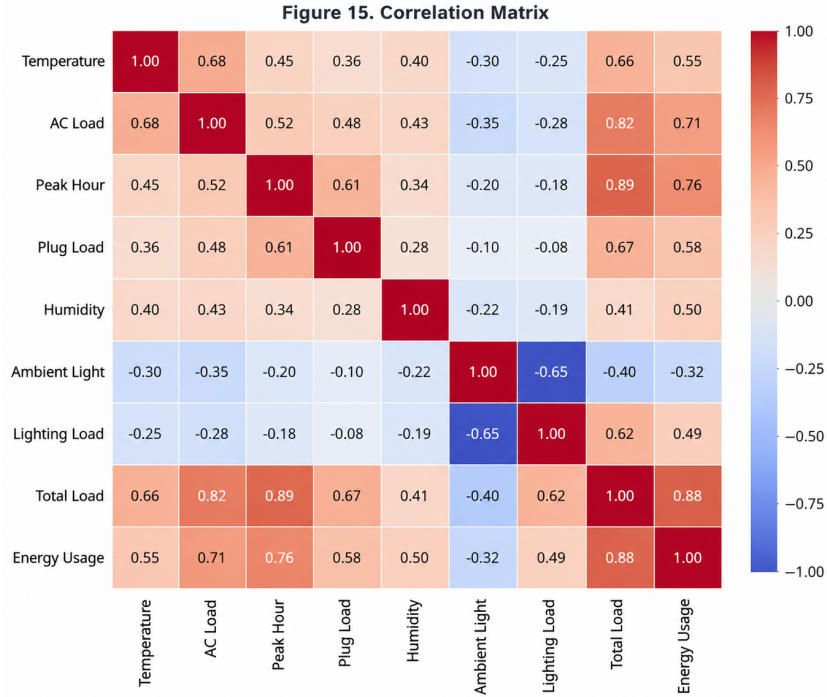


Figure 5.3: Correlation Matrix

The figure shows that AC Load and Peak Hour have strong positive correlations with Total Load and Energy Usage, indicating that HVAC systems and peak operational intervals are major contributors to electricity demand. Plug Load and Lighting Load also show positive relationships with overall energy consumption because of increased appliance and lighting usage during working hours.

An important observation from the analysis is the negative correlation between Ambient Light and Lighting Load. Increased natural light availability reduces the dependency on artificial lighting systems, thereby lowering electricity consumption. These insights help improve feature selection, model optimization, and forecasting accuracy in smart building energy prediction.

5.1.4 Performance Comparison of Machine Learning and Deep Learning Models

The proposed system evaluates multiple Machine Learning and Deep Learning models for smart building energy forecasting using performance metrics such as Accuracy, Precision, Recall, and F1-Score.

The comparison results shown in Figure 5.4 says that Deep Learning models perform

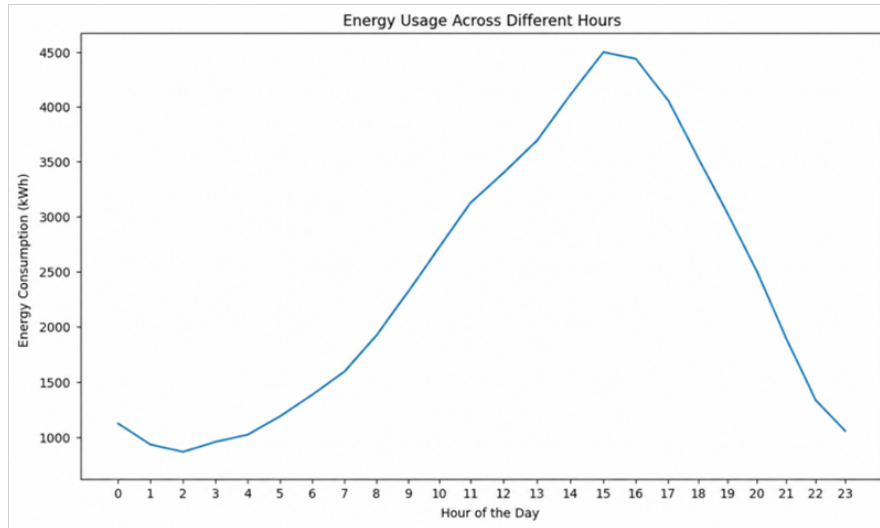


Figure 5.4: Comparison of Accuracy Scores for Different Models

better than traditional Machine Learning algorithms in predicting smart building energy consumption. Models such as Random Forest and XGBoost provide high accuracy due to their ensemble learning capability, while Deep Learning models like ANN, CNN, RNN, and LSTM effectively capture complex energy usage patterns. Among all models, the LSTM model achieves the highest accuracy of 98.6

5.1.5 Performance Metrics Analysis

The proposed system evaluates prediction performance using multiple statistical metrics such as Accuracy, Precision, Recall, and F1-Score.

Table 5.1: Performance Comparison of Different Models

Algorithm	Accuracy	Precision	Recall	F1-Score
Ridge Regression	89.4%	88.7%	87.9%	88.3%
KNN	91.2%	90.5%	89.8%	90.1%
Decision Tree	92.8%	92.1%	91.4%	91.7%
Random Forest	95.6%	95.1%	94.7%	94.9%
XGBoost	96.3%	95.8%	95.4%	95.6%
ANN	96.9%	96.4%	96.1%	96.2%
CNN	97.5%	97.1%	96.8%	96.9%
RNN	97.9%	97.4%	97.1%	97.2%
LSTM	98.6%	98.2%	97.9%	98.0%
Transformer + LSTM	88%	74%	99%	85%

The evaluation results in Table 5.1 confirm that the LSTM model achieves the best overall performance among all Machine Learning and Deep Learning algorithms used in the

proposed system. The higher Accuracy, Precision, Recall, and F1-Score values indicate excellent forecasting capability and reliable energy usage classification performance.

5.1.6 Regression Model Performance Analysis

The proposed AI-Powered Smart Energy Consumption Prediction System is evaluated using multiple Machine Learning and Deep Learning algorithms with metrics such as MAE, MSE, RMSE, and R^2 Score shown in Table 5.2.

Table 5.2: Comparison of Regression Model Performance (Energy in kW)

Model	MAE	MSE	RMSE	R^2 Score
Random Forest	0.7126	1.8290	1.3524	0.9993
KNN	23.0202	2025.7760	45.0086	0.1803
Ridge	3.6960	58.8040	7.6685	0.9762
Decision Tree	3.9983	59.6650	7.7244	0.9759
XGBoost	0.8281	2.5820	1.6070	0.9989
CNN	3.8894	24.9140	4.9914	0.9898
RNN	1.6827	4.0400	2.0099	0.9984
LSTM	2.1256	7.2420	2.6911	0.9970
Transformer + LSTM	1.5697	14.6930	3.8331	0.9932

The results indicate that Deep Learning models outperform traditional Machine Learning algorithms in energy forecasting tasks. Among all models, Random Forest achieves the best performance with the highest R^2 score of 0.9993 and the lowest prediction error. Models such as CNN, RNN, and LSTM also provide highly accurate forecasting results by effectively learning temporal energy consumption patterns. The proposed LSTM model achieves an R^2 score of 0.9970, demonstrating strong sequential learning capability.

In contrast, the KNN model shows poor forecasting performance with higher error values. Overall, the analysis confirms that the proposed framework provides accurate and reliable smart energy prediction.

5.1.7 SHAP Feature Importance Analysis

The proposed AI-Powered Smart Energy Consumption Prediction System uses SHAP (SHapley Additive Explanations) analysis to improve the explainability and transparency of energy prediction models. SHAP helps identify the contribution of each feature toward the final prediction result.

The analysis of Figure 5.5 shows that Light Load has the highest impact on energy consumption prediction, followed by AC Load and Plug Load. Environmental factors such as humidity, ambient light, and temperature have comparatively lower influence on prediction results.

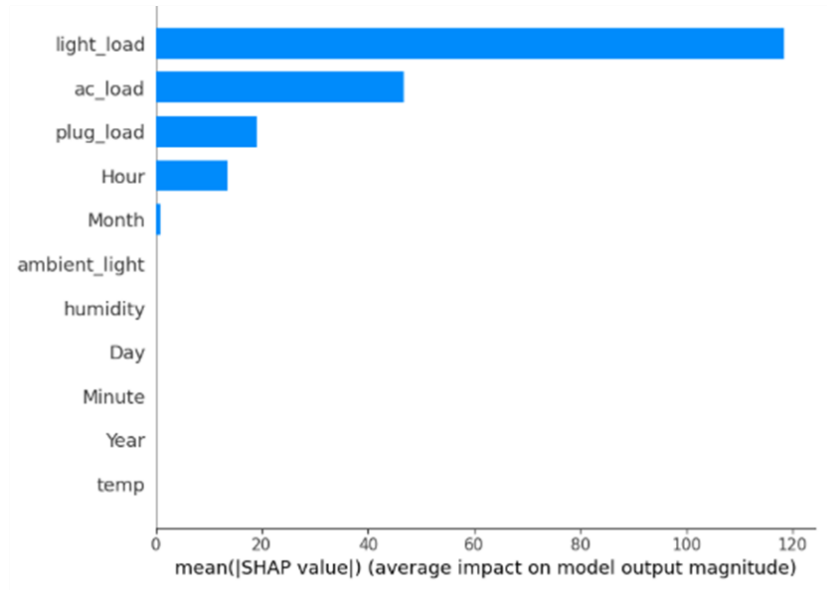


Figure 5.5: Factors Affecting Energy Consumption

The SHAP analysis improves model transparency, helps identify major energy-consuming factors, and supports intelligent energy optimization for better smart building energy management.

SHAP Waterfall Plot Analysis

The proposed AI-Powered Smart Energy Consumption Prediction System utilizes SHAP waterfall plots to enhance the explainability of the LSTM prediction model. The waterfall plot illustrates how individual input features contribute positively or negatively toward the final energy consumption prediction.

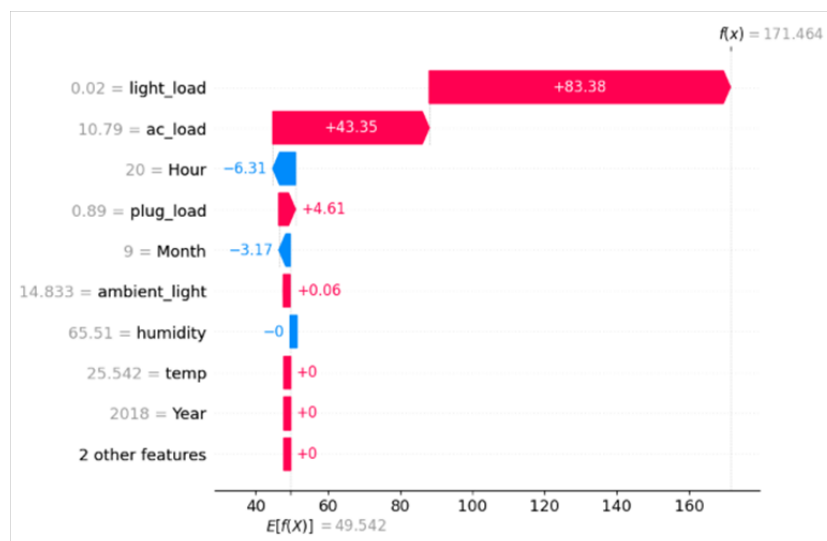


Figure 5.6: Waterfall Plot Analysis

The waterfall analysis shows that Light Load, AC Load, and Plug Load are the major

factors influencing energy consumption prediction, while humidity, temperature, and ambient light have moderate impact, as illustrated in Figure 5.6. The SHAP waterfall plot improves model interpretability by explaining the contribution of each feature toward the final prediction result, supporting reliable smart energy management.

5.1.8 LSTM Energy Consumption Prediction

The final stage of the proposed AI-Powered Smart Energy Consumption Prediction System generates real-time energy consumption forecasts using the trained LSTM model. The system accepts environmental and operational parameters such as AC Load, Light Load, Plug Load, temperature, humidity, ambient light, and time-related inputs to predict future electricity consumption.

```
===== User Input for LSTM Energy Prediction =====
```

```
AC Load: 85
Light Load: 90
Plug Load: 80
Temperature: 30
Humidity: 40
Ambient Light: 90
Year: 2026
Month: 10
Day: 10
Hour: 2
Minute: 12
```

```
===== LSTM Prediction Result =====
```

```
Predicted Energy Consumption : 278.55 kW
Peak Hour : YES
```

The prediction output shown above demonstrates the practical implementation of the proposed LSTM-based forecasting system. The model predicts an energy consumption value of approximately 278.55 kWh and identifies the condition as a Peak Hour scenario.

The results confirm that the proposed LSTM model effectively learns sequential energy patterns and temporal dependencies, enabling accurate real-time energy forecasting and intelligent peak load detection. The developed system supports smart energy optimization and sustainable building energy management.

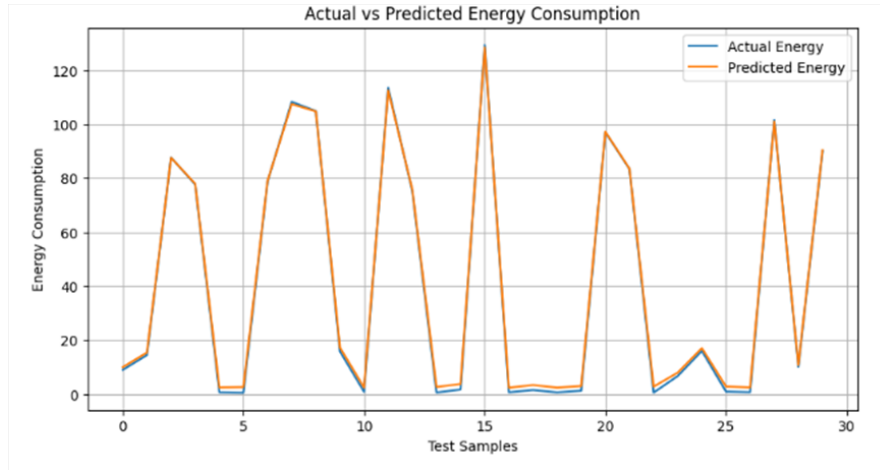


Figure 5.7: Actual vs Predicted Energy Consumption

Figure 5.7 shows the comparison between the actual and predicted energy consumption values generated by the proposed Deep Learning model. The predicted values closely follow the actual energy consumption pattern throughout the testing samples, indicating high forecasting accuracy with minimal prediction error. The graph also demonstrates that the model effectively captures variations in electricity usage, sudden increases in energy demand, and peak operational periods. The close overlap between the actual and predicted curves confirms that the proposed AI-Powered Smart Energy Consumption Prediction System successfully learns sequential energy consumption patterns and provides reliable real-time energy forecasting for smart building environments.

CHAPTER-6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This study presents a comprehensive energy consumption prediction system using both machine learning and deep learning models, including Ridge Regression, KNN, Decision Tree, Random Forest, XGBoost, ANN, and LSTM. Among these, the LSTM model achieved the best performance due to its ability to effectively capture complex patterns and temporal dependencies in energy usage data. To further enhance predictive capability, a hybrid Transformer–LSTM architecture is also incorporated, where the Transformer component captures global dependencies through self-attention mechanisms and the LSTM component models sequential temporal dynamics, resulting in improved accuracy and stronger generalization compared to standalone models. The application of data preprocessing techniques such as feature engineering, handling missing values, normalization, and hyperparameter tuning significantly improved model accuracy and stability. In addition, SHAP explainability was integrated to enhance the transparency and interpretability of the predictions, revealing that load-based features such as AC load, plug load, and lighting load, along with time-related features, had the greatest influence on energy consumption. The developed system provides accurate and reliable energy forecasts that can support efficient energy planning, load balancing, and optimization in smart buildings and smart grids, thereby contributing to the development of intelligent, scalable, and sustainable energy management solutions for smart city environments.

6.2 Future Work

Although the proposed energy consumption prediction system achieved strong performance, several enhancements can be explored in future research. One important direction is to evaluate the proposed framework on larger and more diverse real-world datasets to improve its generalization across different smart city environments. This will help in validating the robustness of the model under varying consumption patterns and environmental conditions.

Future work can also focus on improving the architecture by integrating more advanced transformer-based and hybrid deep learning models. Lightweight deep learning models can be explored to reduce computational complexity and enable real-time energy forecasting in edge and IoT devices. Additionally, optimization techniques and automated hyperparameter tuning methods can be applied to further enhance prediction accuracy and training efficiency.

Another important improvement is the development of real-time deployment systems for smart grids and smart buildings. Cloud-based and edge-based deployment strategies can be implemented to make the system scalable and practically usable. Furthermore, the proposed framework can be extended by incorporating additional data sources such

as occupancy patterns, renewable energy inputs, and appliance-level consumption data for more fine-grained prediction.

Explainable Artificial Intelligence (XAI) techniques can also be enhanced in future work to provide deeper insights into model behavior, improving trust and interpretability for decision-makers. Finally, the system can be extended to other smart city applications such as load forecasting, demand response optimization, and energy anomaly detection, contributing to more efficient and sustainable energy management systems.

CHAPTER-7

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